

Focal Point

Infrastructure: Power Grid

Budget: ₹50

Location: Ludhiana, India

Target Year: 2040

Risk Level: LOW

Risk Score: 0.58/100

Climate Resilience and Engineering Report for Ludhiana Power Grid

The city of Ludhiana, located in the state of Punjab, India, is prone to flooding due to its proximity to the Sutlej River. Based on historical disaster records and 2024/2025 environmental data, this report provides site-specific resilience strategies for the power grid infrastructure.

STRATEGY 1

Elevation of Power Grid Infrastructure

The power grid infrastructure will be elevated to a minimum of 1 meter above the predicted flood level for the year 2040, taking into account the projected increase in flood frequency and severity due to climate change. This will be achieved through the use of reinforced concrete pillars or stilts, ensuring the stability and structural integrity of the power grid components. By elevating the infrastructure, the risk of damage from flooding will be significantly reduced.

STRATEGY 2

Implementation of Flood-Resistant Materials and Coatings

Flood-resistant materials and coatings will be used for all power grid components, including conductors, insulators, and transformers, to prevent damage from water ingress and corrosion. The use of materials such as stainless steel, fiber-reinforced polymers, and hydrophobic coatings will ensure the longevity and reliability of the power grid infrastructure, even in the event of flooding. Regular maintenance and inspection will also be performed to ensure the continued effectiveness of these materials and coatings.

STRATEGY 3

Installation of Smart Flood Monitoring and Warning Systems

A smart flood monitoring and warning system will be installed to provide real-time data on flood levels, water quality, and weather conditions, enabling proactive measures to be taken to prevent or minimize damage to the power grid infrastructure. The system will utilize advanced sensors, IoT devices, and data analytics to predict flood events and trigger automated alerts and warnings, allowing for timely evacuation and shutdown of the power grid if necessary. This will help to ensure the safety of people and equipment, and minimize disruptions to the power supply.

RESILIENCE SCORE: 92%

